

ANSI | ELECTRICAL SAFETY

ANSI/ESD S20.20-2021: Protection of Electrical and Electronic Parts

By Brad Kelechava March 9, 2022



There's electricity within you. The human body, at rest, contains around 100 volts. Our bodies are not alone in generating a charge. In manufacturing and production environments, integrated circuits can generate a charge around 200 volts. These two values comprise the human body model (HBM) and the charged device model (CDM). Their comprehension is essential for protecting electrostatic discharge (ESD) susceptible equipment through an ESD control program in accordance with [ANSI/ESD S20.20-2021: Protection Of Electrical And Electronic Parts, Assemblies And Equipment \(Excluding Electrically Initiated Explosive Devices\)](#).

What is Electrostatic Discharge?

According to the [ESD Association](#), the ANSI-accredited standards developing organization that publishes [ANSI/ESD S20.20-2021, electrostatic discharge \(ESD\)](#) is:

“the rapid, spontaneous transfer of electrostatic charge induced by a high electrostatic field. Note: Usually, the charge flows through a spark between two conductive bodies at different electrostatic potentials as they approach one another”

Definition from [ESD ADV1.0-2017: ESD Association Advisory For Electrostatic Discharge Terminology – Glossary](#).

Many have experienced this phenomenon with a well-known nuisance—when you shuffle across a carpeted floor and touch a doorknob, you feel a startling shock.

ESDs have long been a problem in industry. As far back as the 1400s, European and Caribbean forts limited static to prevent inadvertent ignition of gunpowder stores. By the time of the Industrial Revolution, paper mills throughout the U.S. employed techniques to dissipate static electricity from the paper web as it moved along the drying process.

Electronics have heightened the concerns surrounding electrostatic discharges, since ESDs can be destructive to equipment and components.

What Devices Are Susceptible to ESDs?

Electrostatic discharge susceptible (ESDS) items include, but are not limited to, microcircuits, discrete semiconductors, thick and thin film resistors, hybrid devices, printed circuit boards, and piezoelectric crystals.

Preventing Electrostatic Discharge

ESDs are prevented through grounding and the use of static dissipative materials. Three fundamental control principles form the basis of the [ANSI/ESD S20.20-2021](#) document:

- All conductors in the environment, including personnel, shall be bonded or electrically connected and attached to a known ground or contrived ground.
- Process essential insulators in the environment cannot lose their electrostatic charge by attachment to ground.
- Transportation of ESDS items necessitates enclosures in protective materials, although the type of material depends on the situation and destination.

What is ANSI/ESD S20.20-2021?

An American National Standard, [ANSI/ESD S20.20-2021](#) provides administrative and technical requirements for establishing, implementing, and maintaining an ESD control program.

Such a program addresses:

- Training
- Product qualification
- Compliance verification
- Grounding/equipotential bonding systems
- Personnel grounding
- ESD protected area (EPA) requirements
- Packaging
- Marking

Many of the technical requirements call for test methods outlined in other ESD Association standards. For instance, the grounding/bonding system requirement is tested with [ANSI/ESD S6.1-2019](#). In addition, a [wrist strap system](#) is tested with [ANSI/ESD S1.1-2021](#) (for product qualification) and [ESD TR53-01-18](#) (for compliance verification).

[ANSI/ESD S20.20-2021](#) applies to organizations that manufacture, process, assemble, install, package, label, service, test, inspect, transport, or otherwise handle electrical or electronic parts, assemblies, and equipment susceptible to damage by electrostatic discharges greater than or equal to 100 volts human body model (HBM) and 200 volts charged device model (CDM). Furthermore, this standard handles protection from isolated conductors by limiting the voltage on them to less than 35 volts.

For more information on this standard, there is a technical report, [ESD TR20.20-2016: Handbook For The Development Of An Electrostatic Discharge Control Program For The Protection Of Electronic Parts, Assemblies, And Equipment](#), which you can read more about here: [Handbook for the Development of an Electrostatic Discharge Control Program](#).

What Are the Changes to ANSI/ESD S20.20-2021?

[ANSI/ESD S20.20-2021](#) revises the 2014 edition of the same American National Standard. In addition to editorial changes and some general revisions to improve clarity, it went through the following changes:

- Definitions were added for ESD control items, insulator, conductor, isolated conductor, and unprotected ESDS item.
- Wording for tailoring was updated.
- Update made to allow qualification for ESD control items that stay on-site to be done at the lowest relative humidity (RH) on the site.
- Qualification records were updated to require including supporting technical reports.
- Explanation was added that the flooring/footwear system cannot use compliance verification data for qualification.
- Note was added to the compliance verification plan that calibration does not imply the equipment can make measurements.
- A statement that compliance verification of the grounding system is not required was added.
- The insulator section was updated with a field measurement of where the ESDS item is handled.
- For isolated conductors, the inclusion of non-contact electrostatic voltmeters and electrostatic field meters were included with a note on measurement issues.
- ANSI/ESD STM4.2 was deleted as a qualification for worksurfaces in Table 3, "EPA ESD Control Items." This table also added point-to-point requirements to groundable static control garments and groundable static control garment systems.
- Table 4, "Packaging Requirements," was added. This table summarizes [ANSI/ESD S541-2019: Packaging Materials For The Protection Of Electrostatic Discharge Susceptible Items](#) requirements.
- New Section 8.4.1, "United States Department of Defense (DoD) Packaging Requirements," was added.
- New Informative Annex C, "Tailoring," was added.
- New Informative Annex E, "Statement of Direction," was added.

[ANSI/ESD S20.20-2021: Protection Of Electrical And Electronic Parts, Assemblies And Equipment \(Excluding Electrically Initiated Explosive Devices\)](#) is available on the ANSI Webstore. Any past users of this standard might be interested in [ANSI/ESD S20.20-2021 \(incl. Redline version\)](#), which comes with both the 2021 American National Standard and its Redline version—a document that marks in red every change from the 2014 edition.

Questions About ANSI/ESD S20.20?

Please direct any technical questions relating to this American National Standard to the developer. You can find the contact information for all ANSI-accredited standards developers here: [List of ANSI-Accredited Standards Developers \(ASDs\)](#).

For further information, please refer to: [Who to Contact for Standards Related Questions](#).

Learn More About ANSI/ESD S20.20



Understanding the ANSI/ESD S20.20 Standard for ESD Protection
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January 21, 2024 at 10:16 pm

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Reply



Oscar Armando Ortiz Patiño says:
September 25, 2025 at 11:49 am

Requiero una cotización para la expedición del certificado en norma ANSI/ESD S20.20 para el laboratorio forense y almacén de archivo de evidencia de policía judicial adscrito a la Contraloría General de la Republica sede principal Bogotá Colombia.

Reply



Brad Kelechava says:
October 1, 2025 at 2:13 pm

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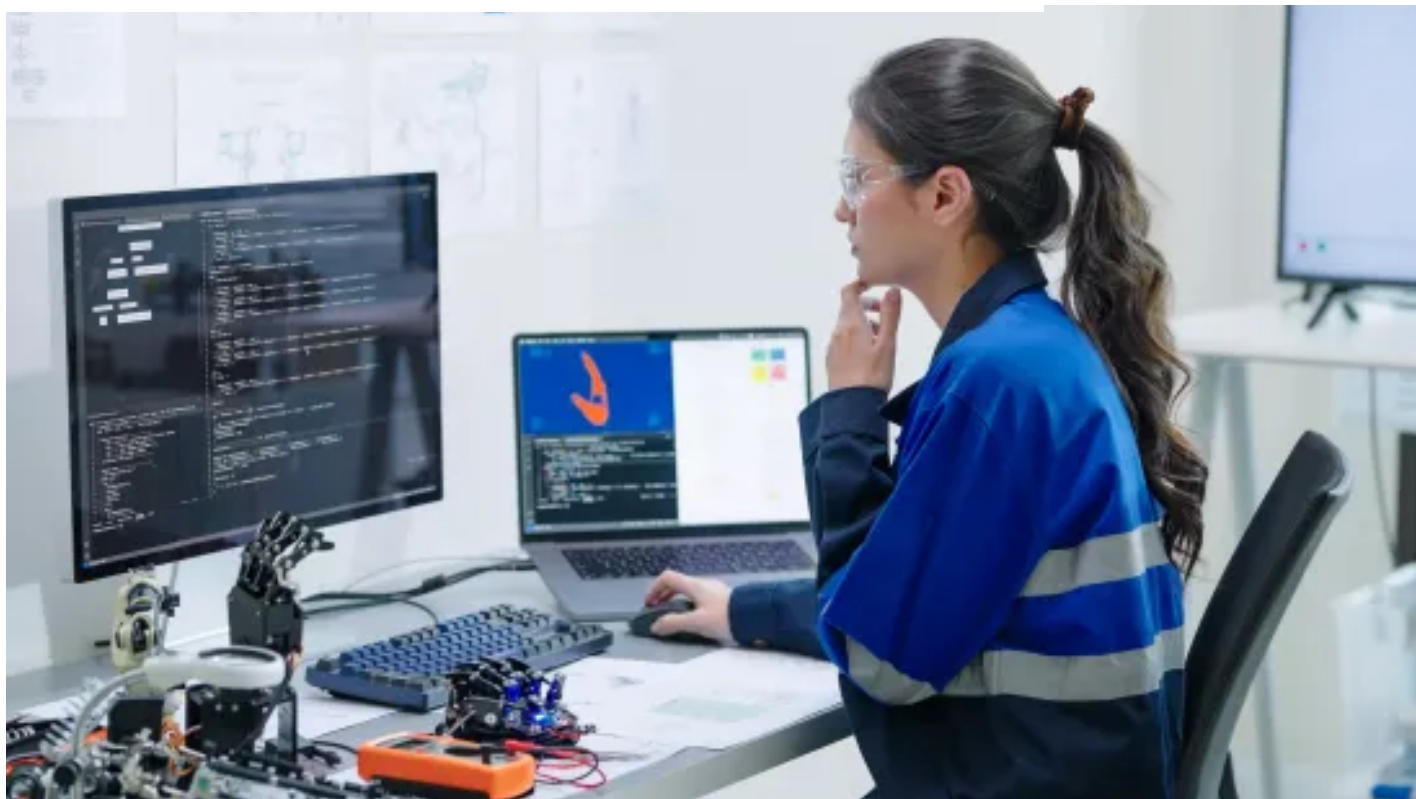
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